Type MS · Drawing the graph from your table

Answer 1 of 15 · Question 7 part b · 2 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

Number			
7 (a)	x | 0 | 0.5 | 1 | 1.5 | 2 | 2.5 | 3 | 3.5 y | 4.26 | **4.14** | 4 | **3.83** | 3.63 | **3.37** | 3 | 2.37	B2 (2)
(b)	Points plotted within half a square. Points joined with a smooth curve.	B1ft B1ft (2)
(c)	$3^{2x-5} = (4-x)^3 \Rightarrow 2x-5 = 3\log_3(4-x)$	M1M1
	$\frac{2x-5}{3} + 3 = \log_3(4-x) + 3 \Rightarrow \frac{2x+4}{3} = \log_3(4-x) + 3$	M1A1
	ALT	
	$\log_3(4-x) + 3 = mx + c$	
	$\Rightarrow \log_3(4-x) = mx + (c-3)$	
	$\Rightarrow (4-x) = 3^{mx+(c-3)}$	[M1
	$\Rightarrow (4-x)^3 = 3^{(mx+(c-3))3} = 3^{3mx+3c-9}$	M1
	$\Rightarrow 2x-5 = 3mx+3c-9 \Rightarrow 3m = 2$ and $-5 = 3c-9$	M1
	$\Rightarrow m = \frac{2}{3} \quad c = \frac{4}{3}$	A1]
	Graph of $y = \frac{2x+4}{3}$ drawn. Intersection point is $x = 2.8$	M1A1 (6) **[10]**

Type MS · Drawing the graph from your table

Answer 1 of 15 · Question 7 part b · 2 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

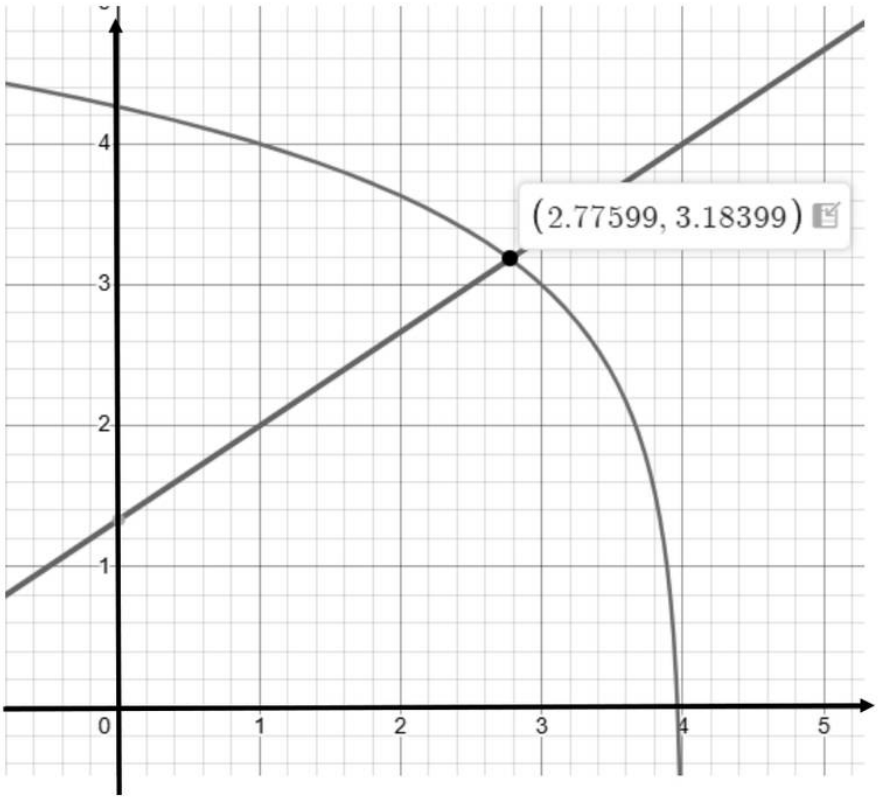
Part	Mark	Additional Guidance
(a)	B2	All three points correct (B1 for 2 points correct)
(b)	B1ft	Points plotted ft their table. Allow half a square tolerance.
	B1ft	Points joined together with a smooth curve. Allow half a square tolerance on joining up their points. FT their table
(c)	M1	For taking logs base 3 of both sides $2x-5=\log_3(4-x)^3$
	M1	For applying the power law $2x-5=3\log_3(4-x)$
	ddM1	For attempting to rearrange to the required form An attempt is defined as dividing through by 3 and adding 3 to both sides.
	A1	For the correct line, in any form $\frac{2x-5}{3}+3=\log_3(4-x)+3\Rightarrow\left[(y=)\frac{2}{3}x+\frac{4}{3}\right]$
	ALT for first 4 marks	
	M1	Sets up the equation to obtain $\log_3(4-x)+3=mx+c$ And removes the log to obtain $(4-x)=3^{mx+(c-3)}$
	M1	Raises **both** sides to a power of 3. Allow one error in this on the LHS
	M1	Compares coefficients to obtain an equation in m and an equation in c
	A1	Obtains the values $m=\frac{2}{3}$ $c=\frac{4}{3}$
	Draws the line	
	M1	For their line drawn provided it is of the form $y=kx\pm\frac{4}{3}$ for $k\neq 0,\ 1$
	A1	$x=2.8$ or 2.7 Calculator value is 2.776

Type MS · Drawing the graph from your table

Answer 1 of 15 · Question 7 part b · 2 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

USEFUL GRAPH



Type MS · Drawing the graph from your table

Answer 1 of 15 · Question 7 part b · 2 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 2 of 15 · Question 8 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
8 (a)	<table><tr><td>x</td><td>0</td><td>0.25</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td><td>3</td></tr><tr><td>y</td><td>2</td><td>**2.41**</td><td>**2.69**</td><td>3.10</td><td>3.39</td><td>3.61</td><td>**3.95**</td></tr></table>	B2 [2]
(b)	Points plotted within half of a square Points joined together in a smooth curve	B1ft B1ft [2]
(c)	$\ln(2x+1) = 3x-4 \Rightarrow \ln(2x+1)+2 = 3x-2$ Graph of $y = 3x-2$ drawn. Intersection point is at $x = 1.8$ or 1.9 (Accept either)	M1 M1A1 [3]
(d)	$e^{(6-x)} = (2x+1)^2 \Rightarrow 6-x = \ln(2x+1)^2 \Rightarrow 6-x = 2\ln(2x+1)$ $\Rightarrow \ln(2x+1)+2 = 5-\frac{x}{2}$ Graph of $y = 5-\frac{x}{2}$ drawn. Intersection point is at $x = 2.4$ or 2.5 (Accept either)	M1 M1 M1A1 [4]
		Total 11 marks

Type MS · Drawing the graph from your table

Answer 2 of 15 · Question 8 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 3 of 15 · Question 7 part b · 2 marks

2023 JAN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

7 a	<table><tr><td>-6</td><td>-5</td><td>-4</td><td>–3</td><td>–2</td><td>–1</td><td>0</td></tr><tr><td>4</td><td>3.59</td><td>3.26</td><td>**3**</td><td>**2.79**</td><td>**2.63**</td><td>2.5</td></tr></table>	B2 (2)
b	Points plotted within a half square Joined with a smooth curve	B1 ft B1 ft (2)
c	$\left(\log_2(2x+2)^3 + x + 3 = 0\right)$	
	$3\log_2(2x+2) = -x-3$ or $\log_2(2x+2) = \frac{-x}{3} - 1$	M1
	$\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = -\frac{x}{3} - 1$ or $2x+2 = 2^{-\frac{x}{3}-1}$ or $\frac{1}{2x+2} = 2^{\frac{x}{3}+1}$	M1
	$\left(\log_{0.5}(2x+2) = \frac{x}{3}+1 \Rightarrow\right) 2x+2 = 0.5^{\frac{x}{3}+1}$	M1
	$2x+4 = 0.5^{\left(\frac{x}{3}+1\right)} + 2$	A1
	$y = 2x+4$ drawn	M1
	Intersect at $x = -0.8 / -0.7$	A1 (6)
ALT	$3\log_2(2x+2) = -x-3$ or $\log_2(2x+2) = \frac{-x}{3} - 1$	M1
	$\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = -\frac{x}{3} - 1 \left(\Rightarrow \log_{0.5}(2x+2) = \frac{x}{3}+1\right)$	M1
	$\left(y-2 = 0.5^{\left(\frac{x}{3}+1\right)} \Rightarrow\right) \log_{0.5}(y-2) = \frac{x}{3}+1 \Rightarrow \log_{0.5}(y-2) = \log_{0.5}(2x+2)$	M1
	$y-2 = 2x+2$	A1
	As main scheme	M1 A1
		Total 10 marks

Type MS · Drawing the graph from your table

Answer 3 of 15 · Question 7 part b · 2 marks

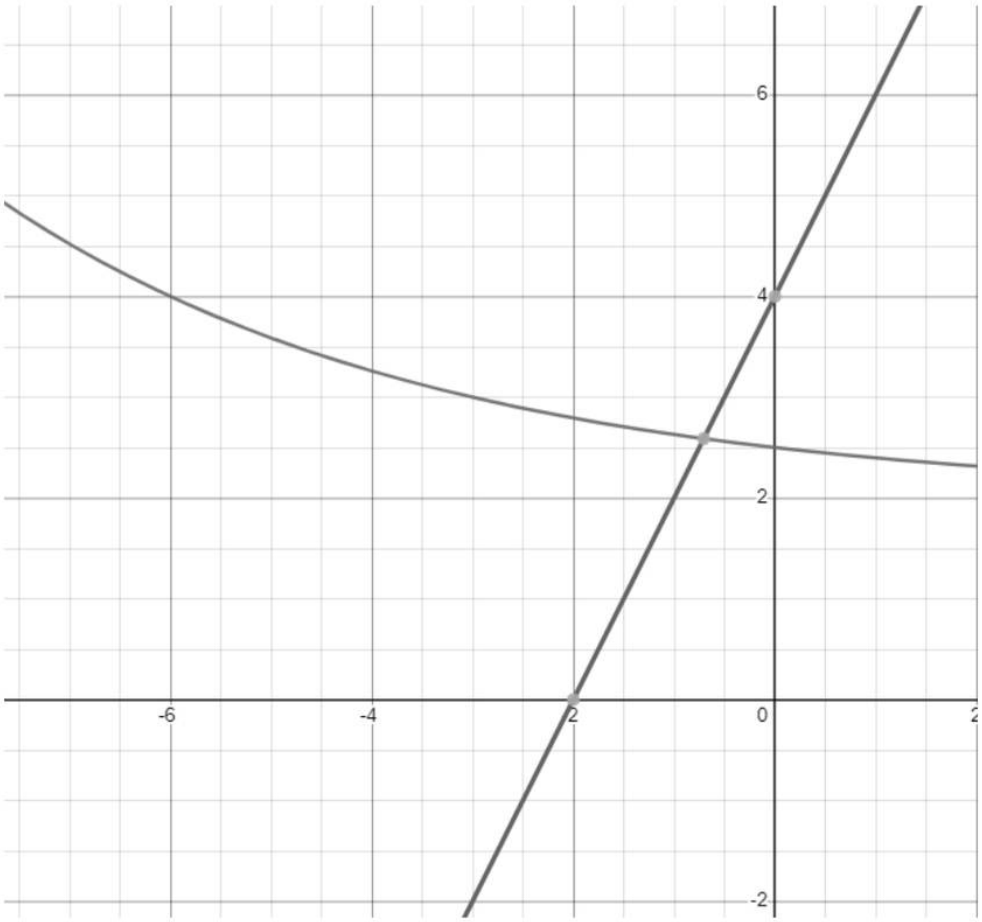
2023 JAN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B2	For all 3 values in the table correct to 2 d.p. Allow 3.0 or 3.00 for "3" (B1 for 2 values)
(b)	B1 ft	For all of the points plotted within half a square, allow use of their values. Points must be checked carefully, including using the zoom tool on ePen if necessary.
	B1 ft	For all of their points joined with a smooth curve. The curve must pass through each of their points to within half a square. Be cautious to not award this mark if straight lines are drawn between the points plotted.
(c)	M1	For use of $\log_a x^k = k\log_a x$ and an attempt to rearrange. Accept an equation of the form $\pm 3\log_2(2x+2) = \pm x \pm 3$ or $\pm\log_2(2x+2) = \frac{\pm x}{3} \pm 1$
	M1	For correctly dividing by 3 (if not already done) and correct use of $\log_b a = \frac{\log_{0.5}(a)}{\log_{0.5}(b)}$ with their equation to get an expression of the form $\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = \pm\frac{x}{3} \pm 1$ or For correctly dividing by 3 (if not done) and correctly converting their equation to exponential form get an equation of the form $2x+2 = 2^{\pm\frac{x}{3}\pm 1}$ or $\frac{1}{2x+2} = 2^{\pm\frac{x}{3}\pm 1}$
	M1	For reaching an equation of the form $2x+2 = 0.5^{\pm\frac{x}{3}\pm 1}$
	A1	For $2x+4 = 0.5^{\left(\frac{x}{3}+1\right)} + 2$
	M1	For $y = 2x+4$ drawn. Correct line drawn can imply any of the 4 previous marks. Allow any line of $y = 2x + c$ to gain this mark. **If the candidate draws any line where c is other than 4, they must have gained the previous 3 method marks.**
	A1	For $x = -0.8 / -0.7$
ALT	M1	For use of $\log_a x^k = k\log_a x$ and an attempt to rearrange. Accept an equation of the form $\pm 3\log_2(2x+2) = \pm x \pm 3$ or $\pm\log_2(2x+2) = \frac{\pm x}{3} \pm 1$
	M1	For correctly dividing by 3 (if not already done) and correct use of $\log_b a = \frac{\log_{0.5}(a)}{\log_{0.5}(b)}$ with their equation to get an expression of the form $\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = \pm\frac{x}{3} \pm 1$
	M1	For an attempt to rearrange and converting to log form to reach $\log_{0.5}(y \pm 2) = \pm\frac{x}{3} \pm 1 \Rightarrow \log_{0.5}(y \pm 2) = \pm\log_{0.5}(2x+2)$
	A1	For the correct equation show.
	M1 A1	As main scheme.

Type MS · Drawing the graph from your table

Answer 3 of 15 · Question 7 part b · 2 marks

2023 JAN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract



Type MS · Drawing the graph from your table

Answer 3 of 15 · Question 7 part b · 2 marks

2023 JAN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 4 of 15 · Question 4 part b · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

4 (a)	x 0.5 1 1.5 2 2.5 3 3.5 y 10 5 4.89 5.5 6.32 7.22 8.16	B2 (2)
(b)	Drawn	B1ft B1ft (2)
(c)	$2x+1+\frac{2}{x^2}=8$ $x=0.6 \quad x=3.4$	M1 A1 (2)
(d)	$2x+1+\frac{2}{x^2}=\frac{1}{2}x+6$ $y=\frac{1}{2}x+6$ drawn $x=0.7 \quad x=3.2$	M1 A1 M1 A1 A1 (5) **[11]**

	Notes
(a)	
B2	For all 4 values correct (B1 for at least 2 values correct)
(b)	
B1ft	For points plotted ft their table (allow half square tolerance)
B1ft	For points joined together with a smooth curve ft their table
(c)	
M1	For $2x+1+\frac{2}{x^2}=8$ may be implied by [y =] 8 identified on the graph
A1	For 0.6 and 3.4 (Allow 0.7 for 0.6)
(d)	
M1	For adding $\frac{1}{2}x$ to both sides of the equation or adding 1 to both sides of the equation
A1	For $2x+1+\frac{2}{x^2}=\frac{1}{2}x+6$
M1	For $y=\frac{1}{2}x+6$ drawn
A1	For 0.7 (allow 0.8)
A1	For 3.2 (allow 3.3)

Type MS · Drawing the graph from your table

Answer 4 of 15 · Question 4 part b · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 5 of 15 · Question 8 part b · 2 marks

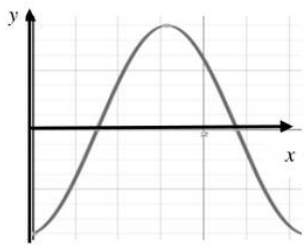
2022 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

8(a)	<table><tr><td>x</td><td>0</td><td>0.5</td><td>0.8</td><td>1</td><td>1.6</td><td>2</td><td>2.5</td><td>3</td></tr><tr><td>y</td><td>**–3.5**</td><td>**–1.6**</td><td>0.3</td><td>1.6</td><td>**3.5**</td><td>**2.4**</td><td>–0.8</td><td>–3.2</td></tr></table>	B2
(b)	Their points plotted correctly on the graph Points are joined to form a smooth curve	B1ft B1ft [2]
(c)	$\cos(A+B)=\cos A\cos B-\sin A\sin B \Rightarrow \cos 2A=\cos^2 A-\sin^2 A=1-\sin^2 A-\sin^2 A$ $\cos(2A)=1-2\sin^2 A$	M1 A1 cso [2]
(d)	$2\sin x+6(1-\cos 2x)-x-5=0$ $\sin x-3\cos 2x-\frac{1}{2}=\frac{x}{2}\pm k$ where k is a constant $\Rightarrow y=\frac{x}{2}-1$ Draws line $y=\frac{x}{2}-1$ $x=0.6$ or $x=0.7$ **AND** $x=2.3$ or $x=2.4$	M1 dM1A1 M1 A1 [5]
	Total	**11 marks**

Type MS · Drawing the graph from your table

Answer 5 of 15 · Question 8 part b · 2 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Part</th><th>Mark</th><th>Scheme</th></tr>
<tr><td>(a)</td><td>B2</td><td>
<table>
<tr><td>x</td><td>0</td><td>0.5</td><td>0.8</td><td>1</td><td>1.6</td><td>2</td><td>2.5</td><td>3</td></tr>
<tr><td>y</td><td>–3.5</td><td>–1.6</td><td>0.3</td><td>1.6</td><td>3.5</td><td>2.4</td><td>–0.8</td><td>–3.2</td></tr>
</table>
For all 4 values correct OR If B2 not scored then award B1 for any 2 values correct
Note: These values must be rounded to one decimal place only. Accept only the above values.
</td></tr>
<tr><td>(b)</td><td>B1ft
B1ft</td><td>


All points plotted correctly (ft their values) on the graph within half of one square.
All of their points are joined to form a smooth curve within half of a square of a point. There must be at least six points plotted and joined.
</td></tr>
<tr><td rowspan="2">(c)</td><td>M1</td><td>For using the identity $\cos(A+B)=\cos A\cos B-\sin A\sin B=\cos^2 A-\sin^2 A$ correctly and the identity $\sin^2 A+\cos^2 A=1$ also correctly.</td></tr>
<tr><td>A1
cso</td><td>For simplifying to the required identity $\cos 2A=1-2\sin^2 A$ *
This is a given result so there must be no errors in their solution.</td></tr>
<tr><td rowspan="5">(d)</td><td>M1</td><td>$2\sin^2 x=1-\cos 2x$
For substituting the above into the given f (x) with no errors.
e.g., $2\sin x+6(1-\cos 2x)-x-5=0$</td></tr>
<tr><td>dM1</td><td>For reaching the equation of the curve on one side and the equation of a straight line on the other.
$\sin x-3\cos 2x-\frac{1}{2}=\frac{x}{2}\pm k$ where k is a constant o.e
Note: This mark is dependent on the previous M mark.</td></tr>
<tr><td>A1</td><td>For the correct equation of the equation of the straight line required $y=\frac{x}{2}-1$</td></tr>
<tr><td>M1</td><td>For drawing their line which must be in the form $y=\frac{x}{2}\pm k$
Correct coordinates for $y=\frac{x}{2}-1$ are; $(0,\ -1),\ \left(1,\ -\frac{1}{2}\right),\ (2,\ 0),\ \left(3,\ \frac{1}{2}\right)$</td></tr>
<tr><td>A1</td><td>For both correct roots of $x=0.6$ or $x=0.7$ AND $x=2.3$ or $x=2.4$</td></tr>
</table>

Type MS · Drawing the graph from your table

Answer 5 of 15 · Question 8 part b · 2 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Mark

Type MS · Drawing the graph from your table

Answer 6 of 15 · Question 6 part b · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

6 (a)	<table><tr><td>x</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>−11.5</td><td>**-2**</td><td>**0.2**</td><td>1.3</td><td>2.7</td><td>**3.8**</td><td>**4.9**</td><td>5.9</td></tr></table>	B2 [2]
(b)	Points plotted within half of a square Points joined together in a smooth curve	B1ft B1ft [2]
(c)	$\frac{x^3-3}{x^2} = ax+b \Rightarrow x^3-3 = ax^3+bx^2 \Rightarrow 0 = x^3(a-1)+bx^2+3$ Comparing coefficients $x^3(a-1)-bx^2+3 = 2x^3+6x^2+3$ $\Rightarrow a=3, b=-6$ so line required is $y=3x-6$ Draws the line $y=3x-6$ and identifies two intersections with the curve when $x=0.8/0.9$ and $x=2.8/2.9$	M1 M1A1 M1 A1 (both) [5]
	ALT	
	$2x^3-6x^2+3=0 \Rightarrow 2x-6=-\frac{3}{x^2}$	{M1}
	$3x-6=x-\frac{3}{x^2}$ so line required is $y=3x-6$	{M1}{A1}
	Draws the line $y=3x-6$ and identifies two intersections with the curve when $x=0.8/0.9$ and $x=2.8/2.9$	{M1} {A1} both [5]
		Total 9 marks

Type MS · Drawing the graph from your table

Answer 6 of 15 · Question 6 part b · 2 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(a)	
B2	All 4 points correct (**B1** for 3 points correct)
(b)	
B1ft	Points plotted ft their table allow half a square tolerance
B1ft	Points joined together with a smooth curve ft their table
(c)	
M1	Setting $x-\frac{3}{x^2}=ax+b$ and simplifying to $x^3(a-1)+bx^2+3$
M1	Comparing coefficients
A1	Identifying that the line required is $y=3x-6$
M1	$y=3x-6$ drawn intersecting the curve in two places
A1	$x=0.8/0.9$ and $x=2.8/2.9$
ALT	
M1	Subtracting 3 from both sides and dividing by x^2
M1	Adding x to both sides
A1	Identifying that the line required is $y=3x-6$
M1	$y=3x-6$ drawn intersecting the curve in two places
A1	$x=0.8/0.9$ and $x=2.8/2.9$

Type MS · Drawing the graph from your table

Answer 6 of 15 · Question 6 part b · 2 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 7 of 15 · Question 5 part b · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

<table>
<tr><td>5 (a)</td><td><table>
<tr><td>x</td><td>0</td><td>0.25</td><td>0.5</td><td>0.75</td><td>1</td></tr>
<tr><td>y</td><td>0.14</td><td>0.29</td><td>0.61</td><td>1.28</td><td>2.72</td></tr>
</table></td><td>first B1
2 values correct

second B1
all 3 values correct
[2]</td></tr>
<tr><td>(b)</td><td>Each point plotted correctly within the correct small square

Smooth curve through the points</td><td>B1ft

B1ft
[2]</td></tr>
<tr><td>(c)</td><td>$e^{3x-2} = 3 - x$

$y = 3 - x$

0.9</td><td>M1

M1

A1ft
[3]</td></tr>
<tr><td colspan="3">Total 7 marks</td></tr>
</table>

Type MS · Drawing the graph from your table

Answer 7 of 15 · Question 5 part b · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 8 of 15 · Question 8 part b · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

8 (a)

x	0.5	0.75	1	1.5	2	3	4	5
y	10	**3.83**	2	**1.33**	**1.75**	**3.33**	5.19	7.12

B2
[2]

(b)

Graph drawn: All points to within half of a square
All points joined together in a smooth curve

B1ft
B1ft
[2]

(c)

$2x+\frac{3}{x^2}-3=ax+b \Rightarrow 2x^3+3-3x^2=ax^3+bx^2$

$\Rightarrow x^3(2-a)-x^2(3+b)+3\equiv 4x^3-10x^2+3$ M1

$2-a=4 \Rightarrow a=-2$

$3+b=10 \Rightarrow b=7$ M1A1

So line is $y=7-2x$

$x=0.6,\ 2.4$ M1A1

[Accept $x=0.7$ and $x=2.3$] [5]

ALT

$\frac{4x^3-10x^2+3}{x^2}=0 \Rightarrow 4x-10+\frac{3}{x^2}=0$ {M1

$2x+\frac{3}{x^2}=10-2x \Rightarrow 2x+\frac{3}{x^2}-3=7-2x$ so line is $y=7-2x$ M1A1

$x=0.6,\ 2.4$ M1A1

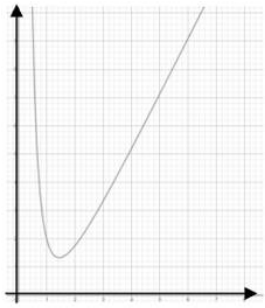
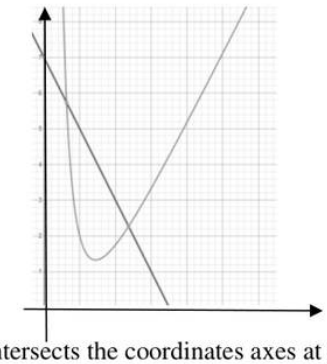
[Accept $x=0.7$ and $x=2.3$] [5]}

Total 9 marks

Type MS · Drawing the graph from your table

Answer 8 of 15 · Question 8 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For two out of 4 values rounded correctly. Penalise rounding only once for awrt the required values.
	B1	For all four values correctly rounded
(b)	B1ft	For **their** points plotted correctly to within half of a square.
	B1ft	For their points joined up in a smooth curve. 
(c)	M1	For setting $2x+\frac{3}{x^2}-3=ax+b$ and attempting to simplify to the form $x^3(2-a)-x^2(3+b)+3$ where the coefficient of each of x^3 and x^2 is in terms of a and/or b
	M1	For equating coefficients and attempting to find the value of a and b
	ALT	
	M1	For dividing through by x^2 to give $4x-10+\frac{3}{x^2}=0$
	M1	For adding 7 and subtracting $2x$ from both sides to give on the LHS $2x+\frac{3}{x^2}-3$
	A1	For the correct straight line $y=7-2x$
	M1	For their line drawn provided it is of the form $y=A-2x$ where $A\neq 0$  The line intersects the coordinates axes at (0, 7) and (3.5, 0)
	A1	For both values of 0.6 and 2.4 Accept 0.7 and 2.3 - without working these roots score no marks. **NB**: Calculator roots are 2.366 and 0.634

Type MS · Drawing the graph from your table

Answer 8 of 15 · Question 8 part b · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 9 of 15 · Question 4 part b · 2 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

4 (a)		

x	0.25	0.5	1	1.5	2	2.5	3
y	**-0.55**	-0.20	-0.30	-0.60	**-0.96**	**-1.35**	**-1.77**

B1

B1

[2]

(b) Points plotted. B1ft

Smooth curve. B1ft

[2]

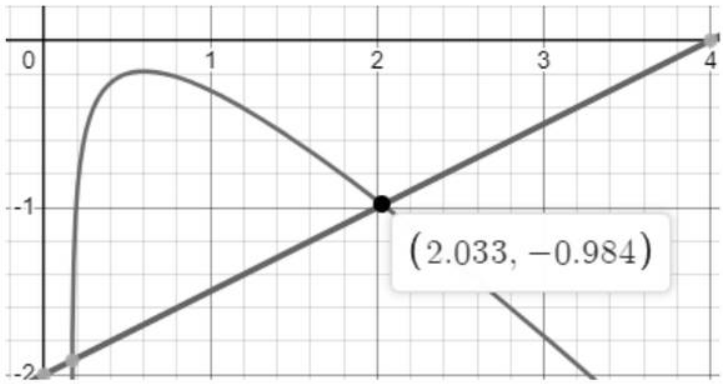
(c) $\frac{3x-4}{2} = \log_{10}(6x-1)$ M1

$\frac{x}{2} - 2 = \log_{10}(6x-1) - x$ M1

Draws the line $y = \frac{x}{2} - 2$ M1

2.0 / 2.1 A1

[4]



Total 8 marks

Type MS · Drawing the graph from your table

Answer 9 of 15 · Question 4 part b · 2 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For 2 values correct. 2dp only.
	B1	For all 4 values correct. 2dp only.
(b)	B1ft	For all 7 points plotted within half a square ft their values from the table.
	B1ft	For points joined with a smooth curve ft all their 7 points plotted, their points do not have be plotted correctly.
(c)	M1	Correctly takes logs to give the expression shown. Accept $\frac{3x-4}{2} = \log(6x-1)$
	M1	Rearranges to give an equation of the form $\pm kx - 2 = \log_{10}(6x-1) - x$ (their left-hand side can be unsimplified equivalent)
	M1	Draws a line of the form $y = \pm kx - 2$ oe, correctly.
	A1	Accept answer given as 2.0 or 2.1
ALT (c)	M1	Correctly rearranges $y = \log_{10}(6x-1) - x$ to $10^{y+x} = 6x-1$
	M1	Equates $y + x$ to $\frac{3x-4}{2}$, and obtains $y = \pm kx - 2$ (can be unsimplified)
	M1	Draws a line of the form $y = \pm kx - 2$ oe, correctly. There must be at least one intersection point with the curve.
	A1	Accept answer given only as either 2.0 or 2.1, Do not allow 2

Type MS · Drawing the graph from your table

Answer 9 of 15 · Question 4 part b · 2 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 10 of 15 · Question 9 part b · 2 marks

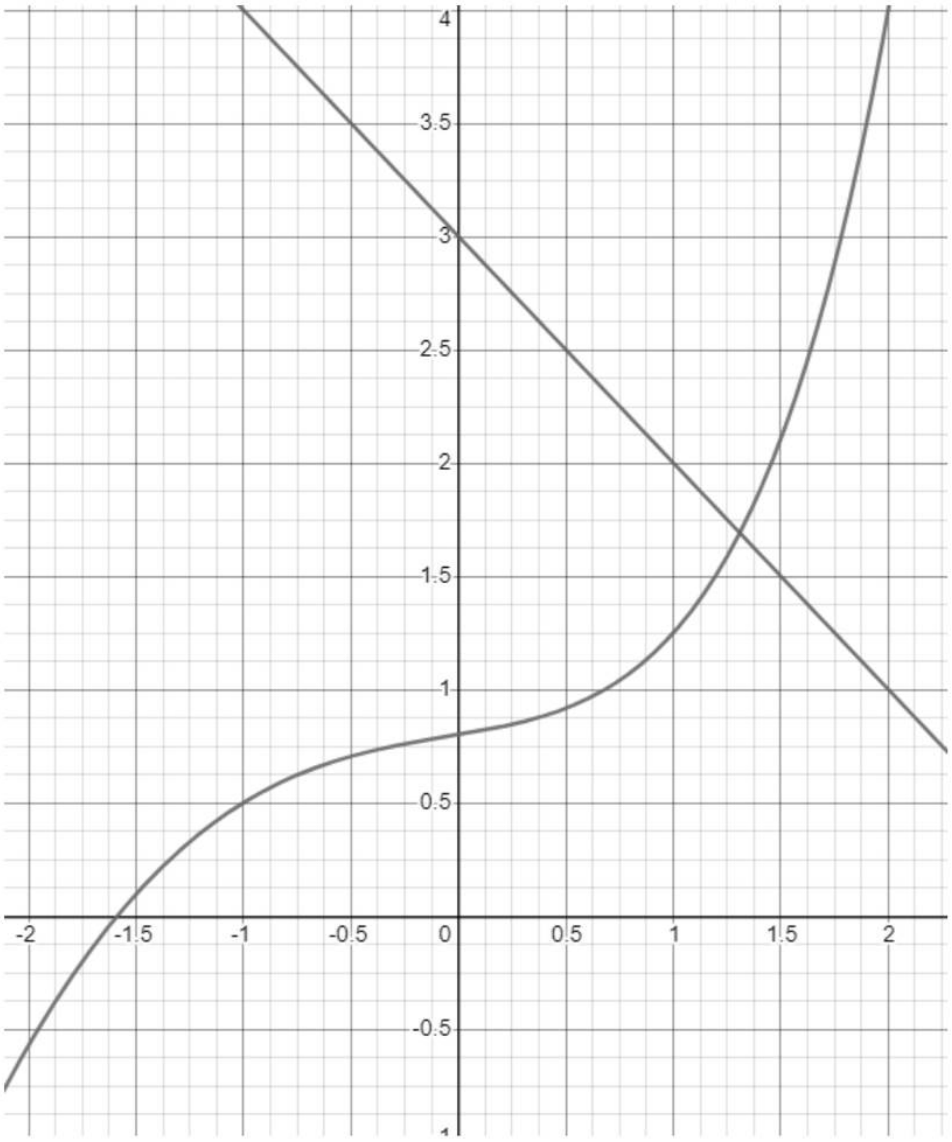
2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

9 (a)	<table><tr><td>–0.5</td><td>0.5</td><td>1.5</td></tr><tr><td>0.70</td><td>0.92</td><td>2.11</td></tr></table>	B1 B1 [2]
(b)	Points plotted	B1ft
	Joined with a smooth curve	B1ft [2]
(c)	$x^3+4=x^2-8x+15$	M1
	$x^3+4=(x-5)(x-3)$ oe	M1
	$\frac{x^3+4}{5-x}=3-x$	A1
	$y=3-x$ drawn	M1
	$x=1.3/1.4$	A1 [5]
ALT	$\frac{x^3+4}{5-x}=ax+b \Leftrightarrow x^3+4=5ax-ax^2+5b-bx$	{M1}
	$\Leftrightarrow x^3+ax^2+\ b-5a\ \ x+4-5b\equiv x^3-x^2+8x-11$ $\Leftrightarrow a=-1\quad b=3$	{M1} {A1}
	$y=3-x$ drawn	{M1}
	$x=1.3/1.4$	{A1} [5]
		Total 9 marks

Type MS · Drawing the graph from your table

Answer 10 of 15 · Question 9 part b · 2 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract



Type MS · Drawing the graph from your table

Answer 10 of 15 · Question 9 part b · 2 marks

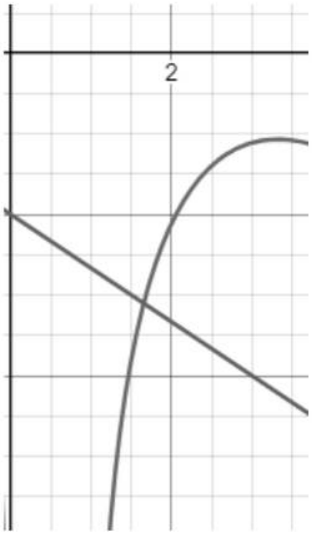
2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 11 of 15 · Question 7 part b · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

7 (a)	<table><tr><td>x</td><td>1.3</td><td>1.5</td><td>2</td><td>2.5</td><td>3</td><td>3.5</td></tr><tr><td>y</td><td>**−5.17**</td><td>−3.79</td><td>−2.11</td><td>**−1.39**</td><td>**−1.11**</td><td>**−1.08**</td></tr></table>	B1 B1 [2]
(b)	Points plotted. Smooth curve.  	B1ft B1ft [2]
(c)	**General principle for marking** M1 – Deal with $9 = 3^2$ or apply addition law and change $\log_3 9 = 2$ M1 – Deal with logs M1 – Apply power law A1 – Correct line $\frac{3^{\frac{4}{3}x}}{3^2} = (x^2 - x)^3$ $3^{\frac{4}{3}x-2} = (x^2 - x)^3$ $\frac{4}{3}x - 2 = \log_3 (x^2 - x)^3$ $\frac{4}{3}x - 2 = 3\log_3 (x^2 - x)$ $-\frac{2}{3}x - 2 = 3\log_3 (x^2 - x) - 2x$	 M1 M1 M1 A1

	OR $3^{\frac{4}{3}x} = 9\left(x^2 - x\right)^3$ $\log_3 3^{\frac{4}{3}x} = \log_3 9\left(x^2 - x\right)^3 \Rightarrow \frac{4}{3}x = \log_3 9\left(x^2 - x\right)^3$ $\frac{4}{3}x = \log_3 9 + \log_3\left(x^2 - x\right)^3 \Rightarrow \frac{4}{3}x = 2 + \log_3\left(x^2 - x\right)^3$ $\frac{4}{3}x - 2 = \log_3\left(x^2 - x\right)^3 \Rightarrow \frac{4}{3}x - 2 = 3\log_3\left(x^2 - x\right)$ $-\frac{2}{3}x - 2 = 3\log_3\left(x^2 - x\right) - 2x$ Draws the line $y = \frac{-2}{3}x - 2$ $(x =)$ 1.6 or 1.7	2nd M1 1st M1 3rd M1 A1 M1 A1 [6]
ALT (c)	Set $3\log_3\left(x^2 - x\right) - 2x = Ax + B$ $\log_3\left(x^2 - x\right)^3 = Ax + 2x + B$ $\Rightarrow \left(x^2 - x\right)^3 = 3^{(A+2)x+B}$ Drawing on given equation $9\left(x^2 - x\right)^3 = 9 \times 3^{(A+2)x+B} = 3^{(A+2)x+B+2}$ Equating exponents $\frac{4}{3}x = (A+2)\text{x} \qquad 0 = B + 2$ $A = \frac{4}{3} - 2 = -\frac{2}{3} \qquad B = -2$ Hence $y = \frac{-2}{3}x - 2$ Draws the line $y = \frac{-2}{3}x - 2$ $(x =)$ 1.6 or 1.7	M1 M1 M1 A1 M1 A1*cso [6]
		Total 10 marks

Type MS · Drawing the graph from your table

Answer 11 of 15 · Question 7 part b · 2 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
7 (a)	B1	For 1 value correct to 2dp
	B1	For all 4 values correct.
(b)	B1ft	For points plotted within half a square ft their values from the table.
	B1ft	For points joined with a smooth curve ft their points plotted.
(c)	**General principles using this method** M1 – Deal with 9 or apply addition law and change $\log_3 9 = 2$ M1 – Deal with logs M1 – Apply power law A1 – Correct line	
	M1	Correct rearrangement and use of index rule to give the equation shown. [Deals correctly with $9 = 3^2$]
	M1	Correctly takes logs with their expression to eliminate the exponential.
	M1	Applies the power law for logs correctly with their expression
	A1	For the equation seen oe
	M1	Draws a line of the form $y = \frac{a}{b}x - 2$ Check the intercept on the y-axis or otherwise (0, –2) (1.5, –3) (3, –4)
	A1*cso	1.6 or 1.7, 1 decimal place only.
ALT (c)	M1	For equating the given formula for y with a generic linear equation and setting out to find the coefficients and uses power law correctly.
	M1	Correctly takes exponential from their expression to eliminate the.log
	M1	Substitutes from their expression into the given equality.
	A1	For finding the coefficients of the linear equation oe
	M1	Draws a line of the form $y = \frac{a}{b}x - 2$ Check the intercept on the y-axis or otherwise (0, –2) (1.5, –3) (3, –4)
	--------- A1*cso	1.6 or 1.7, 1 decimal place only.

Type MS · Drawing the graph from your table

Answer 11 of 15 · Question 7 part b · 2 marks

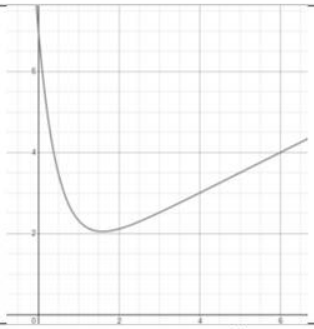
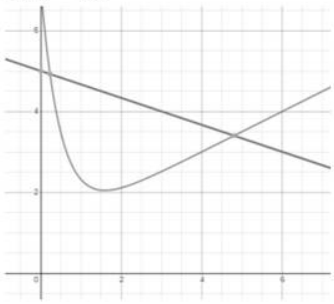
2025 JUN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 12 of 15 · Question 4 part b · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

4(a)	<table><tr><td>x</td><td>0</td><td>1</td><td>1.5</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>7</td><td>**2.3**</td><td>2.0</td><td>**2.1**</td><td>**2.5**</td><td>3</td><td>**3.5**</td><td>4</td></tr></table>	B2 [2]
(b)		B2ft [2]
(c)	$2x+\ln(24-5x)-\ln 36=0 \Rightarrow \ln\left(\frac{24-5x}{36}\right)=-2x$ $\Rightarrow \frac{24-5x}{36}=e^{-2x} \Rightarrow \frac{2}{3}-\frac{5x}{36}=e^{-2x}$ $\Rightarrow 4-\frac{5x}{6}=6e^{-2x}$ $\Rightarrow 5-\frac{x}{3}=\frac{x}{2}+6e^{-2x}+1 \Rightarrow y=5-\frac{x}{3}$ $\Rightarrow x=0.2,\quad 4.8$ 	M1 M1A1 M1A1FT [5]
		Total 9 marks

Type MS · Drawing the graph from your table

Answer 12 of 15 · Question 4 part b · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	At least two correct values rounded correctly.
	B1	All values correct, rounded correctly.
(b)	B1ft	Their points plotted within half of one square.
	B1ft	Their points joined up in a smooth curve. Note: do not allow ruled sections.
(c)	M1	Combines the logs correctly and raises both sides to the power of base e
	M1	Rearranges the equation to obtain the equation of the curve on one side and the equation of the straight line on the other.
	A1	Achieves a correct equation of the straight line. Note: equivalent forms are acceptable.
	M1	Draws their line provided it is of the form $y = k \pm \frac{x}{3}$ when simplified.
	A1FT	Provided M1M1A1M1 awarded. For both correct values of x to 1 decimal place, their values **must** follow through from their graph. 0.2, 0.3, 0.4 and 4.8 Condone coordinates. Note: correct values seen with no working scores M0M0A0M0A0

Type MS · Drawing the graph from your table

Answer 12 of 15 · Question 4 part b · 2 marks

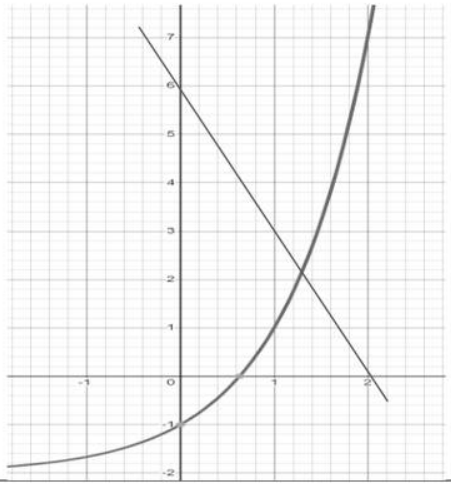
2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 13 of 15 · Question 8 part b · 2 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

8(a)	<table><tr><td>x</td><td>−1</td><td>−0.5</td><td>0</td><td>0.25</td><td>0.5</td><td>0.75</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>y</td><td>−1.7</td><td>−1.4</td><td>−1</td><td>−0.7</td><td>−0.3</td><td>0.3</td><td>1</td><td>3.2</td><td>7</td></tr></table>	B2 [2]
(b)		B2 [2]
(c)	$\log_3 4.5$ $y = 3^x - 2 \Rightarrow 3^x = y + 2 \Rightarrow x = \log_3(y+2)$ $\log_3 4.5 = \log_3(2.5+2) \Rightarrow y = 2.5$ $\Rightarrow x = 1.4$	M1 M1 A1 [3]
(d)	$3^x - 2 = Ax + B \Rightarrow 3^x = Ax + (B+2) \Rightarrow x = \log_3(Ax + (B+2))$ $\Rightarrow \log_3(Ax + (B+2)) - x = 0$ $\log_3(8-3x) - x = 0$ $\Rightarrow A = -3 \quad 8 = B + 2 \Rightarrow B = 6$ $\Rightarrow y = 6 - 3x$ **ALT** $\log_3(8-3x) - x = 0 \Rightarrow \log_3(8-3x) = x$ Taking exponents $8 - 3x = 3^x \Rightarrow 8 - 3x - 2 = 3^x - 2 = y$ $6 - 3x = y$ Draws the line $y = 6 - 3x$ $x = 1.3$	M1 M1 A1 [M1] [M1] [A1] M1 A1 [5]
		Total 12 marks

Type MS · Drawing the graph from your table

Answer 13 of 15 · Question 8 part b · 2 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
8 (a)	B2	B2 for all correct values B1 for a minimum of two values
(b)	B1ft	All points plotted correctly within half of one square.
	B1ft	Their points joined by a smooth curve.
(c)	M1	**In this part, there must be evidence of algebraic work, together with use of the graph.** For using log laws to rearrange the given expression to find $x = \log_3(y+2)$
	M1	For deducing that the required value of y is 2.5 and using their graph to find the value of x at that point.
	A1	For $x = 1.4$
(d)	M1	For setting $3^x - 2 = Ax + B$ and obtaining $x = \log_3(Ax + (B+2))$
	M1	For equating coefficients to deduce that the equation of the line is $y = 6 - 3x$
	A1	For the correct equation of the line. Minimally acceptable equation is $y = k - 3x$ where k is a positive value
	ALT M1	For rearranging the given equation
	ALT M1	For taking exponents and rearranging.
	ALT A1	For the correct equation of the line.
	M1	For drawing their line on the graph. The correct line will cross the axes at (0, 6) and (2, 0)
	A1	For a value of 1.3 **only** The calculator value is 1.291

Type MS · Drawing the graph from your table

Answer 13 of 15 · Question 8 part b · 2 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 14 of 15 · Question 7 part b · 2 marks

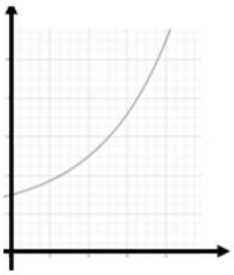
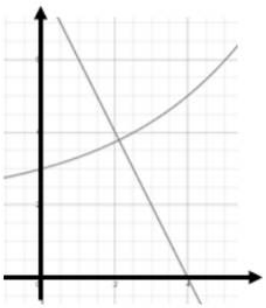
2021 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	<table><tr><td>2</td><td>3</td><td>4</td></tr><tr><td>3.73</td><td>4.28</td><td>5</td></tr></table>	B1 B1 (2)
(b)	Points plotted Joined up with a smooth curve	B1ft B1ft (2)
(c)	$\log_3(6-2x)=\frac{x}{4}$ $6-2x=3^{\frac{x}{4}}$ $8-2x=3^{\frac{x}{4}}+2$ $y=8-2x$ drawn $x=2.1$	M1 M1 A1 M1 A1 (5)
		Total 9 marks

Type MS · Drawing the graph from your table

Answer 14 of 15 · Question 7 part b · 2 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Part</th><th>Mark</th><th>Guidance</th></tr>
<tr><td rowspan="4">(a)</td><td rowspan="2">B1</td><td>For two points (rounded correctly) correct from;
<table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>3</td><td>3.32</td><td>3.73</td><td>4.28</td><td>5</td><td>5.95</td></tr></table></td></tr>
<tr><td>All three points correct and correctly rounded.
Penalise rounding only once here. Condone 5.00</td></tr>
<tr><td rowspan="2">(b)</td><td>B1ft</td><td>All points plotted within half of one square. Ft their values of y for $x = 2,3,4$ respectively</td></tr>
<tr><td>B1ft</td><td>All drawn points joined up in a smooth curve
  </td></tr>
<tr><td rowspan="5">(c)</td><td>M1</td><td>For use of power law to obtain $\log_3(6-2x)=\frac{\pm x}{4}$</td></tr>
<tr><td>M1</td><td>For removing the $\log_3$ to obtain: $6-2x=3^{\frac{\pm x}{4}}$
Allow $(6-2x)^4=3^{\pm x}$ for this mark.</td></tr>
<tr><td>A1</td><td>For obtaining the equation $8-2x=3^{\frac{x}{4}}+2$ oe (eg., $-2x+8=2+3^{\frac{\pi}{4}}$)</td></tr>
<tr><td>M1</td><td>For drawing their straight line, provided it is of the form $y=k-2x$ where k is a constant and $k\neq 6$
 
[Check coordinates (1, 6) (2, 4) (3, 2) (4, 0)]</td></tr>
<tr><td>A1</td><td>For the intersection point $(x=)$ 2.1</td></tr>
</table>

Type MS · Drawing the graph from your table

Answer 14 of 15 · Question 7 part b · 2 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Drawing the graph from your table

Answer 15 of 15 · Question 6 part b · 2 marks

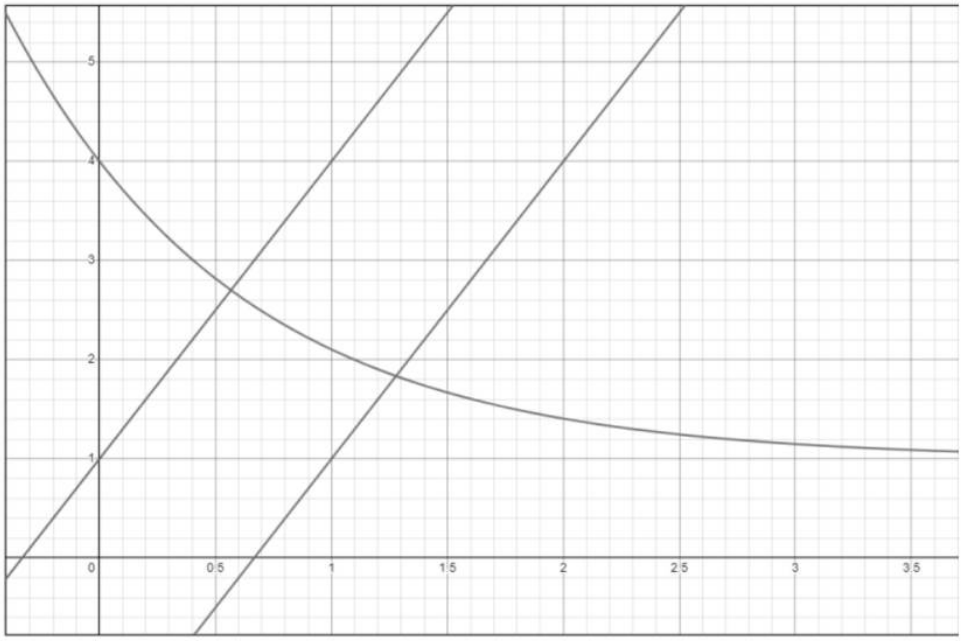
2022 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

6 a	**0** *0.25* *0.5* **1** *1.5* **2** *3* **4(.00)** *3.34* *2.82* **2.1(0)** *1.67* **1.41** *1.15*	B2 (2)
b	Points plotted within half a square Points joined with a smooth curve	B1 ft B1 ft (2)
c	$x = e^{-x} \Rightarrow 3x + 1 = 1 + 3e^{-x}$ or sight of $y = 3x + 1$ and $y = 1 + 3e^{-x}$ $y = 3x + 1$ drawn. Intersection is at $x = 0.5$ or 0.6	M1 M1 A1 (3)
d	$\ln(x-1)^3 = -3x \Rightarrow \ln(x-1) = -x$ $\Rightarrow 3x - 3 = 3e^{-x} \Rightarrow 3x - 2 = 1 + 3e^{-x}$ $y = 3x - 2$ drawn. Intersection is at $x = 1.2$ or 1.3	M1 M1 M1 A1 (4)
ALT – first 2 marks	$(x-1)^3 = e^{-3x} \Rightarrow \left(\sqrt[3]{(x-1)^3} = \sqrt[3]{e^{-3x}}\right) \Rightarrow x - 1 = e^{-x}$ $3x - 3 = 3e^{-x} \Rightarrow 3x - 2 = 1 + 3e^{-x}$	M1 M1
		Total 11 marks

Type MS · Drawing the graph from your table

Answer 15 of 15 · Question 6 part b · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract



Type MS · Drawing the graph from your table

Answer 15 of 15 · Question 6 part b · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
	4 12 19	